Startup Robot

Startup UI:

1. **Computer login**
   1.1. User: Yujie
   1.2. Password: yujie312

2. **Sourcing and building ROS2**
   2.1. In terminal CLI, source ROS2:
   - source /opt/ros/humble/setup.bash

   2.2. Navigate to ros2_ws directory from home directory
   - cd robot_ws

   2.3. Source workspace (build files in ~/ros2_ws/install)
   - source install/setup.bash

3. **Running sawGalilController**
   3.1. Navigate to the json configuration file directory
   - cd ros2_ws/cisst-saw/sawGalilContoller/core/share/test/

   3.2. Start galil_controller and attached json config file. You should see a GUI appear. No need to interact with it.
   - galil_controller -j NPR4.json

   3.3. Verify correction ros topics are being published. We should see relevant CRTK rostopics: measured_cp, measured_js, move_cp, move_js, etc. See CRTK library documentation for more info.
   - ros2 topic list

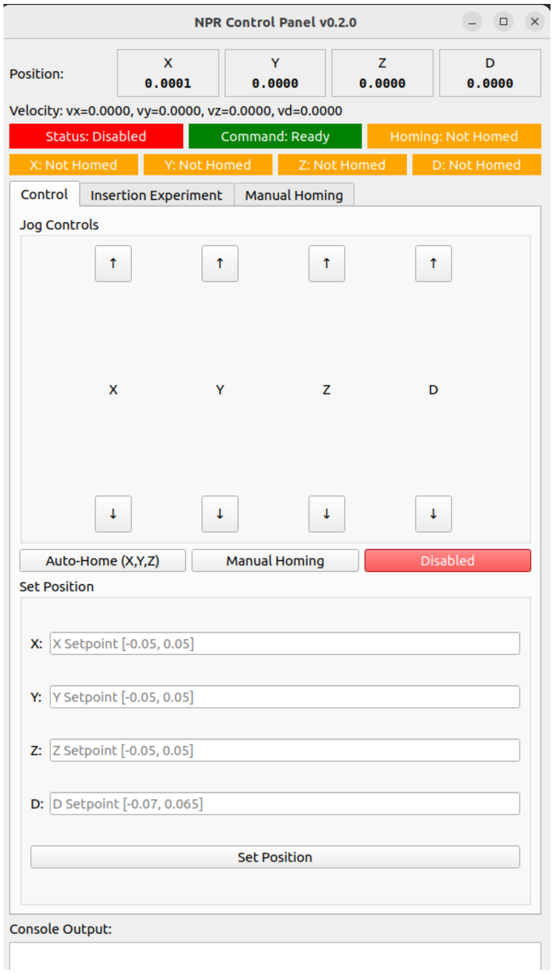
4. **Run the main robot controller GUI script**
   4.1. Repeat step 2 (Sourcing and building ROS2) in a new terminal window.
   4.2. Run GUI and control script. You should see a GUI appear with controls for the robot.
   - ros2 run needlePenetrationRobot insertion_experiment

UI User Guide

The UI has 3 tabs:

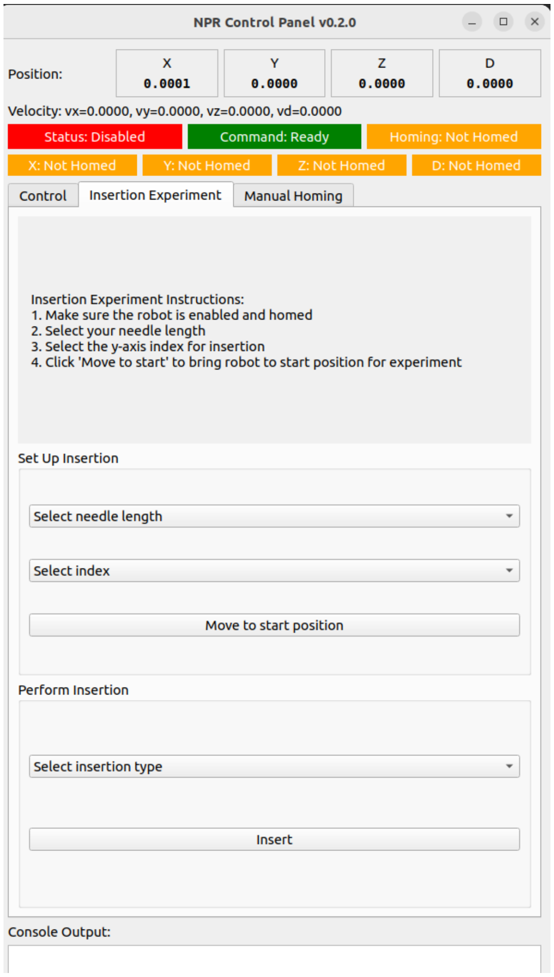
1. Control
2. Insertion Experiment
3. Manual Homing

1. Control



NPR Control Panel v0.2.0
Position:
X 0.0001
Y 0.0000
Z 0.0000
D 0.0000
Velocity: vx=0.0000, vy=0.0000, vz=0.0000, vd=0.0000
Status: Disabled
Command: Ready
Homing: Not Homed
X: Not Homed
Y: Not Homed
Z: Not Homed
D: Not Homed
Control
Insertion Experiment
Manual Homing
Jog Controls
X
Y
Z
D
Auto-Home (X,Y,Z)
Manual Homing
Disabled
Set Position
X: X Setpoint [-0.05, 0.05]
Y: Y Setpoint [-0.05, 0.05]
Z: Z Setpoint [-0.05, 0.05]
D: D Setpoint [-0.07, 0.065]
Set Position
Console Output:

2. Insertion Experiment



3. Manual Homing